

Fear as a Barrier to Coverage: The Chilling Effect of the Public Charge Rule on Health Insurance Enrollment and Healthcare Utilization

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Abstract

Objective: To examine whether the Trump administration’s 2019 expansion of the public charge rule, which penalized immigrants for using Medicaid and other safety-net programs, produced measurable “chilling effects” on health insurance coverage among populations the rule did not legally apply to; specifically, whether any effects persisted after the Biden administration’s 2021 reversal.

Data Sources: American Community Survey, 2016-2024 (N = 5,256,322 U.S. citizen children); California Health Interview Survey, 2015-2024 (N = 220,925 adults).

Study Design: Difference-in-differences and event study models comparing U.S. citizen children in mixed-status families with children in citizen-only families, exploiting state-level variation in noncitizen population share; within-California difference-in-differences comparing noncitizens and naturalized citizens with U.S.-born citizens, with balanced repeated replication variance estimation.

Principal Findings: Medicaid-eligible citizen children in mixed-status families saw significant declines in Medicaid enrollment after the public charge announcement, around 2.0 percentage points ($p < 0.001$) among children below 200% FPL. Pre-trends were flat, the decline peaked at implementation (2020: -2.5 pp, $p = 0.010$), and enrollment has not recovered through 2024 (-3.3

pp, $p < 0.001$). Declines were primarily among children below 200% FPL; placebo tests on citizen-only families showed no effect. In the full sample, uninsurance rose 0.8 percentage points ($p = 0.012$). In California, naturalized citizens, who were formally exempt from the rule, saw a significant 1.3 percentage point decline in Medi-Cal enrollment ($p = 0.030$), consistent with fear-driven spillover. Noncitizens were 20.4 percentage points more likely than citizens to report being asked for Social Security numbers or citizenship documentation when seeking health coverage ($p < 0.001$), which is direct evidence on the bureaucratic mechanism behind chilling effects. All CHIS outcomes pass formal parallel trends tests (all $p > 0.29$).

Conclusions: The public charge rule generated significant chilling effects on health insurance coverage among populations legally exempt from its provisions. These findings show that immigration policy can function as de facto health policy, reducing coverage among eligible citizens and lawful immigrants through fear and administrative barriers rather than through direct eligibility restrictions.

Introduction

In August 2019, the Trump administration finalized a major expansion of the “public charge” inadmissibility rule. For the first time in modern history, receipt of Medicaid, the Supplemental Nutrition Assistance Program (SNAP), and federal housing assistance became negative factors in immigration determinations. While the rule technically applied only to a narrow subset of immigrants seeking admission or adjustment of status and explicitly exempted U.S. citizens, refugees, asylees, and several other categories, its announcement in September 2018 set off widespread fear among immigrant communities that using any government benefit could

jeopardize their family's immigration status (Bernstein, McTarnaghan, and Gonzalez, 2019; Capps et al., 2018).

The resulting "chilling effect" (eligible individuals forgoing benefits they are legally entitled to because of fear, confusion, or misinformation) is among the most consequential unintended consequences of modern immigration policy. The Kaiser Family Foundation estimated that 2.0 to 4.7 million Medicaid and CHIP enrollees could disenroll, including 600,000 to 1.8 million U.S. citizen children who were never subject to the rule (Artiga, Garfield, and Damico, 2019). The Migration Policy Institute documented that safety-net participation among noncitizens declined at twice the rate of citizens during the period around the rule's announcement (Fix and Capps, 2020).

However, the causal evidence on the rule's chilling effects is still fairly limited. The most rigorous study to date, Barofsky et al. (2020), showed that the September 2018 announcement alone reduced children's Medicaid enrollment by approximately 260,000 using county-level administrative data. That study was limited to the announcement period and could not follow the full policy cycle from announcement through implementation, court injunctions, and the eventual reversal under the Biden administration. To my knowledge, no study has used individual-level microdata to estimate causal effects across this complete timeline or to test whether enrollment recovered after the rule was vacated.

In addition, a basic question about mechanisms remains unresolved. Understanding how chilling effects operate requires a nuanced approach that considers the leading hypotheses together - fear of immigration consequences, confusion about the rule's scope, and administrative deterrence through increased documentation demands - because they are hard to separate with enrollment data alone (Perreira and Pedroza, 2019). Direct evidence on whether the bureaucratic process of

seeking coverage became more burdensome for immigrants during this period would substantially improve our understanding of how immigration policy shapes coverage decisions. This paper addresses these gaps with two complementary analyses. The national analysis uses American Community Survey (ACS) data on 5.3 million U.S. citizen children to estimate difference-in-differences (DiD) and event-study models spanning the full public charge policy cycle (2016-2024). The identification strategy compares coverage changes among citizen children in mixed-status families, who are unconditionally eligible for Medicaid but may be deterred by fear of exposing noncitizen parents to immigration scrutiny, with children in citizen-only families, exploiting state-level variation in noncitizen population share as a measure of treatment intensity. The California analysis uses the California Health Interview Survey (CHIS), which can identify noncitizens by documentation status, to directly measure chilling effects on Medi-Cal enrollment and healthcare utilization, including a novel test of the administrative deterrence mechanism.

I add to the literature in several ways: First, it extends the work of Watson (2014) and Alsan and Yang (2024), who established that immigration enforcement, not just eligibility restrictions, generates chilling effects on safety-net participation. To my knowledge, this is the first set of causal estimates of the public charge rule's effects using individual-level microdata across the full policy cycle. Second, it adds to the growing evidence on whether the damage from restrictive policy changes persists after reversal (Bustamante et al., 2022; Gangopadhyaya and Braga, 2022; Hacker, 2004). Lastly, it provides direct evidence on the bureaucratic deterrence mechanism, showing that noncitizens experienced a large increase in documentation demands when seeking coverage - a finding with immediate implications for enrollment outreach and administrative design.

Background

The public charge rule and its timeline

The “public charge” concept dates to the Immigration Act of 1882, which barred admission to persons “likely to become a public charge.” Under the 1999 Field Guidance issued by the Immigration and Naturalization Service, only receipt of cash assistance (Temporary Assistance for Needy Families [TANF] and Supplemental Security Income [SSI]) and government-funded institutional long-term care were considered in public charge determinations. Medicaid, SNAP, and housing assistance were explicitly excluded.

The Trump administration’s proposed rule, announced as a Notice of Proposed Rulemaking (NPRM) in September 2018, sought to expand the definition to include receipt of Medicaid (for non-pregnant adults), SNAP, and federal housing programs for more than 12 months in any 36-month period. The final rule was published in August 2019. The key point for this analysis is that the rule technically applied only to immigrants seeking admission to the United States or adjustment to lawful permanent resident status who were not otherwise exempt. Refugees, asylees, T and U visa holders, VAWA self-petitioners, special immigrant juveniles, and several other categories were explicitly carved out. U.S. citizens, including citizen children of noncitizen parents, were not and could never be subject to public charge determinations.

The implementation timeline created multiple policy shocks that I exploit for identification:

Date	Event
September 2018	NPRM announced; public comment period begins
August 2019	Final rule published
October 2019	Multiple federal courts issue preliminary injunctions blocking implementation
February 2020	Supreme Court stays most injunctions; rule takes effect nationwide except Illinois
July 2020	New injunctions issued during COVID-19

Date	Event
March 2021	pandemic Biden administration stops enforcement; requests court vacatur
March 2022	Biden administration publishes new final rule formally reverting to pre-2019 standards

Despite the rule's narrow legal scope, its chilling effects extended well beyond the directly targeted population: survey evidence from the Urban Institute found that one in five adults in immigrant families with children reported avoiding public benefits due to immigration fears, with 14.8% specifically avoiding Medicaid/CHIP (Bernstein, McTarnaghan, and Gonzalez, 2019). Qualitative evidence from health center staff documented disenrollment among lawfully present immigrants who were fully exempt from the rule (Artiga, Tolbert, and Rudowitz, 2019). This spillover, from a targeted policy to populations it was never designed to affect, is what this paper seeks to causally identify.

Prior evidence on chilling effects

The chilling effects literature begins with the 1996 Personal Responsibility and Work Opportunity Reconciliation Act (PRWORA), which restricted many legal immigrants' access to Medicaid and SNAP. Kandula et al. (2004) showed that PRWORA inadvertently reduced Medicaid enrollment among pre-1996 immigrants who remained fully eligible - a 3 percentage point decline driven by fear and confusion rather than actual eligibility changes. Kaushal and Kaestner (2005) documented even larger effects, with uninsurance among foreign-born women rising 9.9 to 10.7 percentage points.

Watson (2014) provided the seminal causal evidence that immigration enforcement, not just eligibility policy, drives chilling effects. Using instrumental variables, she showed that up to 75% of the decline in noncitizen Medicaid participation around the time of welfare reform was attributable to contemporaneous enforcement spikes. Alsan and Yang (2024) extended this to the

Secure Communities program, showing that its rollout reduced SNAP and ACA enrollment among Hispanic citizen households through network-mediated fear rather than stigma or information barriers.

For the 2019 public charge rule specifically, Barofsky et al. (2020) provided the most causally identified evidence, using county-level difference-in-differences with noncitizen population share as the treatment intensity measure. Their design, which I adapt at the individual level, showed that the announcement alone reduced children's Medicaid enrollment by approximately 260,000. On the utilization side, Haro-Ramos et al. (2026) documented a 2.1 percentage point increase in self-pay emergency department visits among likely undocumented patients at Los Angeles County safety-net hospitals after the announcement.

This study advances the frontier of the aforementioned literature in a number of ways. First, it uses individual-level microdata (ACS, CHIS) rather than county-level aggregates, which allows richer heterogeneity analysis and within-family identification. In addition, it covers the full policy cycle from announcement through reversal, which allows a direct test of whether the effects persisted. Lastly, it exploits CHIS's documentation status variable to study effects on directly identified noncitizens rather than relying on proxy measures, and it provides direct evidence on the administrative deterrence mechanism.

Data

National analysis: ACS and supplemental sources

The national analysis draws on five data sources. The primary individual-level data come from the American Community Survey (ACS), accessed through IPUMS USA, for survey years 2016-2024. The ACS surveys approximately 3.5 million individuals annually, which provides enough

statistical power to identify U.S. citizen children in mixed-status families at the state level. I supplement this with the Current Population Survey Annual Social and Economic Supplement (CPS-ASEC), which offers richer program participation data (including SNAP receipt) at the cost of smaller sample sizes (~95,000 households per year).

For aggregate state-level enrollment trends, I use CMS monthly Medicaid enrollment data from Medicaid.gov, which provides total enrollment by state and month from 2016 to the present.

These data are used for descriptive validation of the individual-level findings, not for the causal estimation itself.

To construct the court injunction instrument, I compiled a federal court injunction timeline from the National Immigration Law Center (NILC) and Catholic Legal Immigration Network (CLINIC) litigation trackers. The timeline documents the exact dates and geographic scope (by federal judicial circuit) of each preliminary and permanent injunction issued against the public charge rule between October 2019 and March 2021.

Lastly, I incorporate state-level policy controls from the Kaiser Family Foundation (Medicaid expansion status and dates), the National Academy for State Health Policy (state immigrant eligibility policies), and the Bureau of Labor Statistics (state unemployment rates), along with ICE enforcement data from the Transactional Records Access Clearinghouse (TRAC) at Syracuse University to control for concurrent immigration enforcement intensity.

Sample construction. The primary analysis sample consists of U.S. citizen children aged 0-18 living in households below 200% of the federal poverty level (FPL) - the income range in which most children are eligible for Medicaid or CHIP. I classify each child into one of two family types using ACS household relationship variables (MOMLOC, POPLOC) and citizenship status (CITIZEN):

- **Mixed-status families:** U.S. citizen children (born in the United States) with at least one parent who is not a U.S. citizen. These children are the “treatment” group. They are unconditionally eligible for Medicaid/CHIP and were explicitly exempt from the public charge rule, so any enrollment changes represent pure chilling effects driven by fear rather than rational responses to actual legal risk.
- **Citizen-only families:** U.S. citizen children with all identified parents holding U.S. citizenship (born or naturalized). These children serve as the within-state control group, as they should be unaffected by the public charge rule.

Treatment intensity. I construct a continuous treatment intensity measure: the state-year noncitizen population share, calculated from the ACS as the proportion of all state residents who report non-citizen status. States with larger noncitizen shares have greater exposure to chilling effects. Mixed-status families are more concentrated in these states, and the immigrant community networks through which fear spreads are denser there (Alsan and Yang, 2024).

Policy timing. I define four policy periods based on the public charge rule timeline:

Period	Dates	ACS Survey Years
Pre-announcement	Before Sep 2018	2016-2018
Announcement/NPRM	Sep 2018 - Feb 2020	2019
Implementation	Feb 2020 - Mar 2021	2020
Reversal	Mar 2021 onward	2021-2024

For the event-study specification, I code year dummies relative to 2018 (the last full pre-announcement year), so the coefficients for each survey year from 2016 to 2024 trace out the dynamic path of enrollment effects.

Outcomes. The primary outcome is an indicator for Medicaid coverage (HINSCAID in the ACS), equal to one if the child is enrolled in Medicaid at the time of the survey. Secondary outcomes include any public health insurance coverage (HCOVPUB), any health insurance coverage (HCOVANY), and private health insurance coverage (HCOVPRIV). In the CPS-

ASEC, I additionally examine SNAP participation (FOODSTMP) as a cross-program spillover outcome.

California analysis: CHIS and ED discharge data

The primary data source for the California analysis is the California Health Interview Survey (CHIS), the largest state-level health survey in the United States. CHIS is particularly valuable for studying immigrant health because it is one of the only population-based surveys that asks respondents about both citizenship and green card holding status. This allows researchers to distinguish among four documentation status groups: (1) U.S.-born citizens, (2) naturalized citizens, (3) permanent residents (green card holders), and (4) non-citizens without green cards - a widely accepted proxy for undocumented status (Pourat et al., 2014; Ortega et al., 2018; Bustamante et al., 2022; Pintor et al., 2025).

I pool CHIS adult public-use files from 2015 through 2024, covering both biennial (2015-2016) and annual (2017 onward) survey waves. The pooled sample yields 220,925 adult respondents, with approximately 16,744 classified as noncitizens. CHIS is administered in English, Spanish, Chinese, Korean, Vietnamese, and Tagalog, which mitigates non-response bias among immigrant populations - an important advantage over surveys conducted only in English and Spanish.

Analysis groups. I define three comparison groups for the within-California difference-in-differences:

Group	CHIS Definition	Rationale
Treatment: Noncitizens (pooled)	CITIZEN2 = “non-citizen”	Most directly affected by public charge fears
Control 1: Naturalized citizens	CITIZEN2 = “naturalized citizen”	Legally exempt from the rule; face some fear but have documentation
Control 2: U.S.-	CITIZEN2 = “US-born”	No public charge risk;

Group	CHIS Definition	Rationale
born citizens		cleanest control
Outcomes. Primary outcomes from CHIS are Medi-Cal enrollment (INSMC), emergency department visit in the past 12 months (ER), having a usual source of care (USUAL), and delayed or foregone medical care (FORGO). I also examine a direct measure of the administrative deterrence mechanism: whether respondents report being asked for a Social Security number or proof of citizenship when seeking health coverage (SSNCIT_YN). ED discharge data. I plan to supplement the CHIS analysis with California emergency department discharge data, to be obtained through a data request to the California Department of Health Care Services (DHCS). The key feature of the ED analysis is the use of Medi-Cal aid codes to directly identify emergency department visits by noncitizens receiving emergency-only (restricted-scope) Medi-Cal coverage. California assigns aid codes 5A/5B (undocumented adults) and 5C/5D (undocumented children) for emergency services. These codes would let me track the monthly volume of emergency Medi-Cal ED visits before and after the public charge announcement, testing whether noncitizens avoided emergency care they were legally entitled to regardless of the rule. I have not yet received these data, so the results below use CHIS only. Data limitations. A few limitations should be noted up front. First, the ACS does not directly identify undocumented individuals. The “mixed-status family” proxy relies on parental non-citizen status, which includes lawfully present non-citizens, and this measurement error likely attenuates the estimates. Second, the CHIS public-use file’s CITIZEN2 variable collapses all non-citizens into a single category, so green card holders cannot be separated from undocumented individuals. I am pursuing CHIS restricted-use data through the UCLA Data Access Center, which provides the underlying variable (AH40) that allows the four-group classification. Third, CHIS sample sizes for noncitizen subgroups are adequate for pooled		

analysis but may be too small for precise year-by-year event-study estimates. I address this by pooling waves and reporting confidence intervals that reflect the statistical uncertainty.

Methods

National analysis (ACS)

Difference-in-differences model. The primary specification is a two-way fixed effects difference-in-differences model:

$$Y_{isrt} = \alpha + \beta_1 \text{Mixed}_i \times \text{Post}_t + \gamma X_{it} + \delta_s + \theta_t + \varepsilon_{isrt}$$

where Y_{isrt} = the health insurance outcome for child i in state s , race/ethnicity group r , in year t , Mixed_i = an indicator for living in a mixed-status family, Post_t = an indicator equal to one for survey years 2019 and later (after the September 2018 NPRM announcement), X_{it} = controls for age, sex, race/ethnicity, and poverty status, δ_s and θ_t = state and year fixed effects, and ε_{isrt} represents an error term. Standard errors are clustered at the state level. Citizen children in mixed-status families are considered as the “treated” group, while children in citizen-only families serve as “untreated” controls. The coefficient of interest, β_1 , captures the differential change in coverage among citizen children in mixed-status families relative to those in citizen-only families after the public charge announcement, net of common time trends and permanent differences between family types. The identifying assumption is that, absent the public charge rule, coverage trends for the two family types would have evolved in parallel. Under that assumption, β_1 estimates the causal chilling effect.

Triple-difference model. To exploit variation in exposure to the rule, I also estimate a triple-difference model:

$$Y_{isrt} = \alpha + \beta_2 \text{Mixed}_i \times \text{HighNC}_s \times \text{Post}_t + [\text{lower-order interactions}] + \gamma X_{it} + \delta_s + \theta_t + \varepsilon_{isrt}$$

where HighNC_s = an indicator for states in the top quartile of noncitizen population share. The triple-interaction coefficient β_2 isolates the additional chilling effect experienced by children in mixed-status families in states with large noncitizen populations, relative to the same comparison in low-noncitizen states. This specification relaxes the parallel trends assumption somewhat: mixed-status and citizen-only families are allowed to have different trends, as long as the difference in trends is constant across high- and low-noncitizen states.

Event studies and COVID-19 adjustment. To test the parallel trends assumption and trace the dynamics of the effect, I estimate an event-study version of the model that interacts the mixed-status indicator with each survey year, with 2018 as the omitted reference year. The coefficients for years before 2018 (the leads) test the parallel trends assumption - they should be close to zero and statistically insignificant. The coefficients for 2019 and later trace the post-announcement trajectory. The COVID-19 pandemic (2020-2021) coincided with the rule's implementation and could confound the estimates through differential economic shocks, Medicaid enrollment changes, or mortality across family types. I address this by 1) dropping survey years 2020-2021 entirely, and 2) including a COVID indicator interacted with treatment and control group status to absorb differential pandemic effects.

California analysis (CHIS)

Basic and multi-group specifications. The basic CHIS specification is a within-California DiD comparing noncitizens to U.S.-born citizens before and after the announcement:

$$Y_{it} = \alpha + \beta_3 \text{Noncitizen}_i \times \text{Post}_t + \gamma X_{it} + \theta_t + \varepsilon_{it}$$

where Y_{it} = the outcome for adult i in year t , Noncitizen_i = an indicator for non-citizen status (pooled in the public-use file), Post_t = an indicator equal to one for survey years 2019 and later, X_{it} = controls for age, sex, race/ethnicity, and poverty status, θ_t = year fixed effects that absorb

California-specific time trends, and ε_{it} represents an error term. Survey years 2015-2018 are considered as the pre-period and 2019-2024 as the post-period. Because CHIS provides a within-state comparison, state-level confounders such as Medi-Cal expansion and economic conditions are differenced out. To test for differential effects across immigration status groups, I estimate a multi-group specification that includes separate post-period interactions for noncitizens and naturalized citizens, with U.S.-born citizens as the omitted reference group:

$$Y_{it} = \alpha + \beta_4 \text{Noncitizen}_i \times \text{Post}_t + \beta_5 \text{Naturalized}_i \times \text{Post}_t + \gamma X_{it} + \theta_t + \varepsilon_{it}$$

The coefficient β_5 on the naturalized citizen interaction is a direct test of chilling spillover.

Naturalized citizens were unambiguously exempt from the public charge rule, so any enrollment decline among this group reflects fear rather than a rational behavioral response.

Administrative deterrence and variance estimation. I test the administrative deterrence mechanism with a regression of the documentation-demand indicator on noncitizen and naturalized indicators plus the same individual controls. The outcome indicates whether the respondent reported being asked for a Social Security number or citizenship documentation when seeking health coverage. Because this variable captures the administrative encounter rather than the enrollment decision, it provides direct evidence on whether the bureaucratic process became more burdensome for immigrants. CHIS uses a complex survey design with stratified random sampling and raked weights, so I implement balanced repeated replication (BRR) variance estimation using the 80 replicate weights provided in the public-use file (RAKEDW0 through RAKEDW80). Each coefficient is re-estimated across all 80 replicate samples, and the BRR standard error is computed from the squared deviations of the replicate estimates around the full-sample estimate.

Results

National analysis (ACS)

Descriptive statistics. In the ACS analysis sample, children in mixed-status families have substantially higher Medicaid enrollment rates (approximately 58.7% vs. 49.9% for citizen-only families; these means are preliminary pending the full restriction of the sample to below 200% FPL), consistent with the higher poverty rates and Medicaid eligibility of immigrant families. Both groups experienced rising Medicaid enrollment over the study period, driven by the ACA and expanded outreach, but the pre-post means suggest a relative flattening for mixed-status families after 2018.

Basic DiD estimates. Table 1 presents the main difference-in-differences estimates for the national analysis. The basic DiD specification yields a coefficient of -0.005 on the Medicaid outcome ($SE = 0.005$, $p = 0.34$) - directionally negative but statistically insignificant. The -0.5 percentage point estimate is economically meaningful, translating to approximately 60,000 children nationally, but the 95% confidence interval $[-1.4, +0.5 \text{ pp}]$ is too wide to rule out either a meaningful decline or no effect.

The results are substantially more precise for the uninsured rate. Citizen children in mixed-status families saw a statistically significant 0.8 percentage point increase in uninsurance ($SE = 0.003$, $p = 0.012$) relative to children in citizen-only families after the announcement. The mirror image shows up for any coverage: a -0.8 percentage point decline ($p = 0.012$). Significant effects on the total coverage margin but not on the specific Medicaid outcome is a pattern consistent with some families moving away from public coverage without successfully obtaining private alternatives.

Triple-DiD estimates. The triple-difference specification, which interacts the mixed-status x post indicator with a high-noncitizen-share state indicator, yields a coefficient of -0.003 on

Medicaid (SE = 0.007, $p = 0.70$). The lack of an additional intensity gradient may reflect countervailing forces. High-noncitizen states have more potentially affected families, but they also tend to have stronger immigrant-serving infrastructure, multilingual outreach, and more welcoming state policies (e.g., California, New York, Illinois), which may partially buffer chilling effects. The mixed-status x high-noncitizen two-way interaction is positive and significant (beta = 0.057, $p < 0.001$), confirming that children in mixed-status families in high-noncitizen states have substantially higher baseline Medicaid enrollment, consistent with these states' more generous eligibility policies.

Event-study estimates. Figure 1 and Table 3 present the event-study coefficients for the Medicaid outcome. The pre-trend coefficients for 2016 (beta = 0.007, $p = 0.34$) and 2017 (beta = 0.005, $p = 0.33$) are small, positive, and statistically insignificant, which supports the parallel trends assumption. After the September 2018 announcement, the trajectory shifts. The 2019 coefficient (the NPRM announcement year) is near zero (beta = -0.002, $p = 0.66$), but the effect deepens substantially at implementation. The 2020 coefficient, the year the rule took effect, is -0.011 ($p = 0.016$), the only post-period coefficient that is individually significant.

The reversal-period coefficients (2021-2024) show a mixed pattern. The 2021 coefficient is positive (beta = 0.003, $p = 0.63$), consistent with partial recovery immediately after the Biden administration vacated the rule. Interestingly, the 2022 coefficient (beta = 0.008, $p = 0.027$) is positive and significant, suggesting over-recovery, potentially reflecting pent-up enrollment or expanded outreach. The 2023 coefficient returns to near zero (beta = 0.004, $p = 0.22$), and the 2024 coefficient is negative and insignificant (beta = -0.006, $p = 0.38$). Overall, the full event-study pattern is consistent with a temporary, acute chilling effect concentrated at rule

implementation that largely dissipated after reversal, although the imprecision of individual year estimates limits strong conclusions about the speed or completeness of recovery.

COVID-19 sensitivity. Table 2 presents the COVID sensitivity analyses. Dropping survey years 2020-2021 reduces the Medicaid DiD estimate modestly (from -0.005 to -0.003, $p = 0.62$) and reduces the sample by 20% ($N = 4,203,852$). Including COVID interaction controls yields nearly identical results ($\beta = -0.003$, $p = 0.60$). These specifications confirm that the estimates are not driven by differential pandemic effects, though they also confirm the limited statistical power for the Medicaid outcome.

California analysis (CHIS)

Descriptive statistics. Table 4 presents CHIS descriptive statistics by citizenship group.

Noncitizens have substantially lower Medi-Cal enrollment (5.7%) compared to naturalized citizens (25.4%) and U.S.-born citizens (22.4%), reflecting ineligibility for full-scope Medi-Cal prior to the 2024 expansion for most noncitizen categories. Noncitizens are younger (mean age 42 vs. 53 for naturalized citizens), more likely to be below 200% FPL (57.0% vs. 36.3% for naturalized, 25.7% for US-born), and substantially less likely to have a usual source of care (71.5% vs. 86.2% for naturalized citizens).

The baseline difference in the documentation demand variable is particularly noteworthy: 11.7% of noncitizens report being asked for a Social Security number or citizenship documentation when seeking coverage, compared to 7.3% of naturalized citizens and 0% of U.S.-born citizens (by construction, as the question is only asked of immigrant respondents).

Basic DiD estimates. Table 6 presents the basic DiD estimates from CHIS. The Medi-Cal enrollment result is null: the noncitizen $\times$ post coefficient is -0.001 ($SE = 0.006$, $p = 0.82$; BRR $SE = 0.001$). This null is expected, as most noncitizens in the pooled public-use sample were

ineligible for full-scope Medi-Cal during the study period. If a chilling effect is present, it would show up mainly as avoidance of Emergency Medi-Cal and other restricted-scope coverage, which the binary Medi-Cal enrollment indicator may not fully capture.

The ED visit and usual source of care outcomes show statistically significant positive coefficients: +5.5 percentage points for ED visits ($p < 0.001$) and +5.5 percentage points for usual source of care ($p < 0.001$). These positive signs are counterintuitive if read as evidence that noncitizens increased healthcare use after the announcement. However, the pre-post means in Table 5 show that these coefficients reflect relative trends. ED visit rates declined for both groups during the study period (from 24.2% to 18.0% for US-born citizens; from 18.0% to 16.1% for noncitizens), but citizens' rates declined faster. The positive DiD coefficient therefore indicates that noncitizens' ED utilization did not fall as quickly as citizens' - consistent with citizens' ED use falling sharply during COVID while noncitizens, who lack alternative care sources, kept using the ED as a primary care setting. The no-COVID specification (dropping 2020-2021) strengthens this coefficient to +6.4 percentage points, which supports the COVID-era compositional explanation.

The forgone care result is null ($\beta = -0.003$, $p = 0.70$), though BRR variance estimation yields a significant estimate (BRR $p = 0.008$). Accounting for the complex survey design therefore reveals a marginally significant decline in reported forgone care among noncitizens relative to citizens - a finding that likely reflects differential reporting rather than genuinely reduced barriers.

Multi-group analysis. Table 7 presents the multi-group specification, which separately estimates post-period changes for noncitizens and naturalized citizens relative to U.S.-born citizens. The key finding is that naturalized citizens saw a significant decline in Medi-Cal

enrollment: the naturalized x post coefficient is -0.013 (SE = 0.006, $p = 0.030$). Naturalized citizens are unconditionally eligible for Medi-Cal and were explicitly exempt from the public charge rule, so this 1.3 percentage point decline is direct evidence of chilling spillover - fear extending beyond the targeted population to immigrants with full legal status.

Naturalized citizens also show a significant decline in forgone care ($\beta = -0.012$, $p = 0.027$).

This may look paradoxical, but it likely reflects that the immigrants who disenrolled from Medi-Cal also reduced their contact with the healthcare system, making them less likely to report forgoing care.

Mechanism: administrative deterrence. Table 8 presents the most novel finding of the California analysis: the documentation demand results. Noncitizens are 20.4 percentage points more likely than U.S.-born citizens to report being asked for a Social Security number or proof of citizenship when seeking health coverage ($p < 0.001$, 95% CI: [19.1, 21.8 pp]). Naturalized citizens experience a similar but smaller disparity: they are 12.2 percentage points more likely to be asked for documentation ($p < 0.001$).

These disparities are large, robust to controls for age, sex, race/ethnicity, and poverty status, and persist in the full CHIS sample ($N = 220,925$). The gradient (largest for noncitizens, intermediate for naturalized citizens, zero for U.S.-born citizens) is consistent with a screening process in which eligibility workers or enrollment assisters apply heightened scrutiny to immigrant-appearing applicants. Being asked for documentation that immigrants may not have (or may fear presenting due to immigration concerns) is a direct bureaucratic barrier to enrollment.

This specification captures a cross-sectional documentation demand disparity rather than a change over time. Even so, the size of the disparity provides direct evidence for the administrative deterrence channel through which chilling effects operate. The public charge rule

raised immigrant communities' awareness that interacting with government agencies could expose their documentation status, and the CHIS data confirm that immigrants who do seek coverage encounter exactly the kind of documentation demands that would deter those fearful of immigration consequences.

This finding extends the framework of Perreira and Pedroza (2019), who hypothesized that immigration policy generates health effects through both formal eligibility restrictions and informal barriers including fear, confusion, and institutional deterrence. To my knowledge, the 20.4 percentage point documentation demand disparity is the first quantification of the institutional deterrence channel.

BRR variance estimates. Appendix Table B confirms that the CHIS estimates hold up under proper survey variance estimation. BRR standard errors are uniformly smaller than robust standard errors, indicating that the conventional robust estimates are conservative. The BRR analysis uses all 80 CHIS replicate weights.

Robustness checks

Below 200% FPL sample. A key concern with the full-sample national estimates is that many citizen children in mixed-status families have incomes above Medicaid/CHIP eligibility thresholds, which dilutes the treatment effect. Restricting the sample to children below 200% FPL (N = 1,622,582; 30.9% of the full sample) yields substantially sharper estimates. Medicaid-eligible children in mixed-status families saw a 2.0 percentage point decline in Medicaid enrollment (SE = 0.005, $p < 0.001$) - four times the magnitude of the full-sample estimate and statistically significant. The any-public-coverage result is nearly identical (-2.0 pp, $p < 0.001$). The event study on this restricted sample (Figure 3) shows clean pre-trends (2016: +0.004, $p = 0.70$; 2017: +0.002, $p = 0.76$), a significant decline at implementation (2020: -2.5 pp, $p = 0.010$),

and enrollment that has not recovered through 2024 (-3.3 pp, $p < 0.001$). This suggests the chilling effect did not fully reverse after the 2021 vacatur and may have compounded as the Medicaid unwinding further disrupted coverage in mixed-status families.

Formal parallel trends tests. Joint F-tests on the pre-period event-study coefficients support the parallel trends assumption for the primary specification. For Medicaid in the full ACS sample, the pre-period F-statistic is 0.946 ($p = 0.39$), failing to reject the null of parallel trends. For the uninsured outcome, the F-statistic is 3.49 ($p = 0.03$), driven by a marginally significant 2016 coefficient (-0.008, $p = 0.019$); the adjacent 2017 coefficient is small and insignificant (-0.003, $p = 0.30$). I view this as a minimal concern for a few reasons. The significant 2016 coefficient is negative (mixed-status children had relatively lower uninsurance in 2016), which is the opposite direction from the post-treatment effect. If anything, this pre-existing pattern biases the estimates toward zero, making the post-period increase in uninsurance a conservative estimate of the true chilling effect. In addition, the 2017 coefficient (the year immediately before treatment) is close to zero and insignificant. Lastly, the preferred specification (the below-200% FPL Medicaid sample) shows clean pre-trends ($F = 0.08$, $p > 0.90$) with both 2016 and 2017 coefficients near zero and insignificant, while yielding a larger and highly significant treatment effect (-2.0 pp, $p < 0.001$).

For the California analysis, all four CHIS outcomes pass the parallel trends test: Medi-Cal ($F = 0.49$, $p = 0.69$), ED visit ($F = 0.56$, $p = 0.64$), usual source of care ($F = 1.24$, $p = 0.29$), and forgone care ($F = 0.64$, $p = 0.59$).

Placebo tests. I conduct two placebo tests for the national analysis. First, estimating the DiD on citizen-only families in high- vs. low-noncitizen-share states yields null coefficients for Medicaid (-0.004, $p = 0.42$), uninsured (+0.002, $p = 0.39$), and any coverage (-0.002, $p = 0.39$), confirming

that citizen-only families experienced no differential coverage change. Second, the placebo event study for citizen-only families x high noncitizen share shows no systematic pattern across years, with all coefficients statistically insignificant (Figure 4).

For the California analysis, the naturalized-citizen placebo shows a significant Medi-Cal decline (-0.013, $p = 0.03$). This is not a “failed” placebo. Rather, it confirms the chilling spillover documented in the multi-group specification: even naturalized citizens, who face zero legal risk from the rule, reduced Medi-Cal enrollment.

Alternative treatment timing. Using implementation (2020) rather than announcement (2019) as the treatment date, the uninsured DiD remains significant (+0.005, $p = 0.035$). The results are not sensitive to the choice of treatment timing.

Heterogeneity. Declines were primarily among Hispanic children in mixed-status families, consistent with the hypothesis that the rule disproportionately affected Latino immigrant communities where fear networks are densest. In the California analysis, Hispanic noncitizens show the largest (though not individually significant) Medi-Cal coefficient, while effects on Asian noncitizens are smaller, consistent with differences in community information networks and documentation status composition. Children below 200% FPL saw substantially larger declines than those above 200% FPL, as expected given that lower-income families depend more on public coverage and have fewer private alternatives.

Covariate balance. Pre-period covariate balance (Figure 5) shows the expected differences between treatment and control groups: mixed-status children have higher Medicaid enrollment (+26.5 pp) and higher poverty rates (+24.8 pp) than citizen-only children, and a 3.4 pp gap in uninsurance. Standardized differences exceed 0.5 for poverty and Medicaid, reflecting basic socioeconomic differences between immigrant and non-immigrant families. These level

differences are absorbed by the family-type fixed effect in the DiD design; the identifying assumption requires only parallel trends, not parallel levels.

Discussion

Summary of findings

Medicaid-eligible citizen children in mixed-status families saw significant declines in Medicaid enrollment after the public charge announcement, around 2.0 percentage points ($p < 0.001$) among children below 200% FPL. Pre-trends were flat, the decline peaked at implementation (2020: -2.5 pp, $p = 0.010$), and enrollment has not recovered through 2024 (-3.3 pp, $p < 0.001$). Declines were primarily among children below 200% FPL; placebo tests on citizen-only families showed no effect. In the full sample, uninsurance rose 0.8 percentage points ($p = 0.012$). In California, naturalized citizens, who were formally exempt from the rule, saw a significant 1.3 percentage point decline in Medi-Cal enrollment ($p = 0.030$), consistent with fear-driven spillover. Noncitizens were 20.4 percentage points more likely than citizens to report being asked for Social Security numbers or citizenship documentation when seeking health coverage ($p < 0.001$), which is direct evidence on the bureaucratic mechanism behind chilling effects.

Comparison to prior literature

The findings are broadly consistent with, but more conservative than, prior estimates. Barofsky et al. (2020) estimated that approximately 260,000 children lost Medicaid coverage nationally using county-level data, while the individual-level estimate here of 0.8 percentage points on uninsurance translates to a similar order of magnitude given the approximately 7.5 million children below 200% FPL in mixed-status families. The gap between Barofsky et al.'s significant Medicaid estimate and the insignificant one here may reflect attenuation from the

individual-level proxy for mixed-status families (which includes some families with lawfully present noncitizen parents who face lower chilling risk) or the greater noise in survey-based coverage measurement relative to administrative enrollment data.

The naturalized citizen Medi-Cal decline (-1.3 pp) is consistent with Urban Institute surveys showing that one in five adults in immigrant families avoided benefits (Bernstein, McTarnaghan, and Gonzalez, 2019) and with Bustamante et al.'s (2022) finding that immigrant families continued avoiding Medi-Cal even after the rule's reversal. The estimate here is smaller than the survey-based reports, which makes sense: survey respondents report intentions to avoid, while the DiD estimates capture actual enrollment changes.

The documentation demand finding (a 20.4 pp disparity) is, to my knowledge, the first population-based quantification of the administrative deterrence channel. Artiga, Tolbert, and Rudowitz (2019) documented qualitative evidence of increased documentation demands at health centers, and Watson (2014) theorized that enforcement creates bureaucratic deterrence, but no prior study has measured the size of documentation disparities at the individual level in a representative sample.

Policy implications

These findings have a few immediate policy implications.

First, immigration policy functions as de facto health policy: the public charge rule, an immigration regulation nominally concerned with visa and green card adjudication, generated measurable declines in health coverage among U.S. citizen children who could never be subject to immigration penalties. Immigration policy changes should therefore be evaluated more holistically, not just for their direct immigration effects but also for their downstream health consequences on populations well outside the policy's formal scope.

Second, administrative design matters as much as formal eligibility. The 20 percentage point documentation demand disparity shows that immigrants face substantial informal barriers when seeking coverage, independent of whether they are legally eligible. Reducing these barriers could reduce chilling effects even when restrictive immigration policies are in place. Examples include language-accessible enrollment assistance, clear messaging that public benefit use does not affect citizenship applications, and training eligibility workers to distinguish between documentation required for enrollment versus documentation relevant to immigration adjudication.

Lastly, the chilling effect operates through fear networks, not just individual rational responses. Naturalized citizens face zero risk from the public charge rule, yet they experienced significant Medi-Cal disenrollment, which suggests that fear spreads through immigrant community networks regardless of individuals' actual legal exposure. This is consistent with Alsan and Yang's (2024) finding that network effects, rather than stigma or information barriers, are the primary transmission channel for enforcement-related chilling effects. Policy responses should therefore target community-level reassurance, not just individual-level eligibility clarification.

Limitations

My study design has several strengths, including two large population-based data sources, a quasi-experimental design that covers the full policy cycle, and direct evidence on the administrative mechanism. Potential limitations include measurement error in the treatment definition, imprecision on the Medicaid margin, and COVID-19 confounding. I want to be direct about them:

- The ACS-based mixed-status family proxy pools lawfully present noncitizens (green card holders, visa holders) with undocumented parents. Lawfully present noncitizens face lower chilling risk than undocumented individuals, so the estimates likely represent a

lower bound on the true chilling effect for families with undocumented members. I do not anticipate that this changes the direction of the findings, as measurement error of this type biases the estimates toward zero. The CHIS restricted-use data, which directly identifies documentation status, will address this attenuation in future analyses.

- The CHIS public-use file pools all noncitizens into a single category, so I cannot test the prediction that chilling effects are strongest among undocumented individuals. The mechanism result (documentation demands) should be read as the average disparity across all noncitizen types. I expect substantially larger effects for the undocumented subgroup once restricted-use data become available.
- The Medicaid enrollment result in the national analysis is imprecisely estimated, with a 95% confidence interval that includes both economically meaningful effects and zero. The significant uninsurance result provides indirect evidence of a coverage decline, but the imprecision on the Medicaid-specific effect limits my ability to quantify the rule's impact on a single program.
- The COVID-19 pandemic coincided with the rule's implementation period, which makes for a difficult identification environment. To address this, I 1) drop the 2020-2021 survey years, and 2) include COVID interaction controls; both suggest that pandemic-related confounding does not drive the main results, but I cannot fully rule out differential pandemic effects on mixed-status families that are correlated with both COVID exposure and chilling.
- The CHIS documentation demand variable captures a cross-sectional disparity rather than a change induced by the public charge rule. I interpret this result as evidence on the mechanism through which chilling effects operate (immigrants encounter documentation

demands that the rule made more threatening), but I cannot establish that the 20.4 percentage point disparity increased specifically because of the 2019 rule.

- Several data sources that would strengthen the analyses are still pending. I plan to 1) obtain CHIS restricted-use data through the UCLA DAC (the application is in preparation), and 2) add CPS-ASEC extracts for cross-program spillover analysis, along with TRAC enforcement data for state-level controls and DHCS ED discharge data for the utilization analysis. The current results are a foundation that will be updated as these data become available.

Future research

I propose a few extensions that would strengthen the evidentiary base of this paper. Restricting the ACS sample to children below 200% FPL (the Medicaid-eligible population) should yield sharper Medicaid-specific estimates. The court injunction DiD, which exploits the geographic variation in federal court injunctions blocking the rule between October 2019 and February 2020, provides a secondary identification strategy independent of the noncitizen share treatment intensity measure. Heterogeneity analyses by race/ethnicity, poverty level, and state policy environment will show which subpopulations experienced the largest chilling effects. Obtaining CHIS restricted-use data will allow the full four-group analysis separating undocumented individuals from green card holders. Lastly, DHCS ED discharge data will allow a test of whether noncitizens shifted from Emergency Medi-Cal to self-pay status, extending Haro-Ramos et al.'s (2026) single-system finding to the statewide level.

Conclusion

The public charge rule of 2019 illustrates how immigration policy can generate far-reaching health consequences beyond its intended scope. Using nationally representative ACS data and California's CHIS survey, this paper provides causal evidence that the rule reduced health insurance coverage among U.S. citizen children and naturalized citizens, populations that were legally exempt from its provisions. It also documents, for the first time to my knowledge, the size of the documentation disparities that immigrants encounter when seeking coverage, quantifying the administrative deterrence channel through which chilling effects operate. These findings arrive at a moment of renewed policy relevance. As immigration enforcement intensifies and new public charge regulations are considered, the results here point to the need for policymakers to anticipate and mitigate the health coverage consequences of immigration policy. An estimated 7.5 million U.S. citizen children live in mixed-status families, and their health coverage should not be collateral damage of immigration enforcement. Protecting eligible populations from chilling effects requires clear federal guidance exempting public benefits from immigration determinations. Alongside this, it requires investment in community-level outreach, multilingual enrollment assistance, and administrative reforms that reduce the documentation burdens deterring eligible immigrants from the coverage they are entitled to.

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Tables

Table 1. ACS Children Change in coverage over time, DiD estimates

Specification	Beta (DiD)	SE	p-value	N
Basic DiD: Medicaid	-0.005	0.005	0.34	5,256,322
Basic DiD: Uninsured	+0.008	0.003	0.012	5,256,322
Basic DiD: Any coverage	-0.008	0.003	0.012	5,256,322
Triple-DiD: Medicaid	-0.003	0.007	0.70	5,256,322

Notes: Dependent variable is an indicator for the named outcome. Treatment group: US citizen children in mixed-status families. Control group: US citizen children in citizen-only families. All specifications include state and year FE, controls for age, sex, race/ethnicity, and poverty. SEs clustered at the state level. Bold indicates $p < 0.05$.

Table 2. ACS Children Change in coverage over time, Medicaid, COVID-19 sensitivity

Specification	Beta (DiD)	SE	p-value	N
No COVID (drop 2020-21)	-0.003	0.006	0.62	4,203,852
COVID interaction control	-0.003	0.006	0.58	5,256,322

Notes: Same specification as the basic DiD in Table 1, with the listed COVID adjustment. SEs clustered at the state level.

Table 3. ACS Children Change in coverage over time, Medicaid, event study

Year (relative to 2018)	Beta	SE	p-value
2016 (-2)	+0.007	0.008	0.34
2017 (-1)	+0.005	0.005	0.33
2018 (ref)	0	-	-
2019 (+1)	-0.002	0.005	0.66
2020 (+2)	-0.011	0.004	0.016
2021 (+3)	+0.003	0.007	0.63
2022 (+4)	+0.008	0.004	0.027
2023 (+5)	+0.004	0.003	0.22
2024 (+6)	-0.006	0.007	0.38

Notes: Coefficients from the mixed-status x year interaction in the event-study specification.

2018 is the omitted reference year. Controls: age, sex, race/ethnicity, poverty, state and year FE.

SEs clustered at the state level. Bold indicates $p < 0.05$.

Table 4. CHIS Adults Descriptive statistics by citizenship group

Variable	US-Born Citizens	Naturalized Citizens	Noncitizens (Pooled)
N	168,525	35,656	16,744
Medi-Cal enrollment	0.224	0.254	0.057
ED visit (past 12 mo)	0.205	0.169	0.169
Usual source of care	0.857	0.862	0.715
Forgone care (past 12 mo)	0.103	0.064	0.079
Asked SSN/citizenship	0.000	0.073	0.117
Age (mean)	46.5	53.3	42.2
Female	0.511	0.524	0.497
Below 200% FPL	0.257	0.363	0.570

Notes: California Health Interview Survey, 2015-2024 pooled. Statistics are unweighted means.

SSN/citizenship question asked only of immigrant respondents.

Table 5. CHIS Adults Change in outcomes over time, pre-post means

Outcome	Period	US-Born	Naturalized	Noncitizens
Medi-Cal	Pre (2015-18)	0.213	0.242	0.056
Medi-Cal	Post (2019-24)	0.231	0.261	0.058
ED visit	Pre (2015-18)	0.242	0.176	0.180
ED visit	Post (2019-24)	0.180	0.164	0.161
Usual source	Pre (2015-18)	0.882	0.880	0.702

Outcome	Period	US-Born	Naturalized	Noncitizens
Usual source	Post (2019-24)	0.841	0.851	0.725
Forgone care	Pre (2015-18)	0.089	0.058	0.068
Forgone care	Post (2019-24)	0.112	0.067	0.087

Notes: Unweighted pre/post means. Pre = 2015-2018; Post = 2019-2024.

Table 6. CHIS Adults Change in outcomes over time, basic DiD

Outcome	Beta (DiD)	Robust SE	p-value	BRR SE	N
Medi-Cal enrollment	-0.001	0.006	0.82	0.001	185,269
ED visit (past 12 mo)	+0.055	0.010	<0.001	0.001	185,269
Usual source of care	+0.055	0.012	<0.001	0.002	185,269
Forgone care	-0.003	0.007	0.70	0.001	185,269

Notes: DiD estimates comparing noncitizens (pooled) vs. US-born citizens before (2015-18) and after (2019-24) the public charge announcement. Controls: age, sex, race/ethnicity, poverty, year FE. BRR SE computed from 80 CHIS replicate weights. N reflects the two-group sample (excludes naturalized citizens for the basic specification).

Table 7. CHIS Adults Change in outcomes over time, multi-group DiD

Outcome	Noncit x Post Beta	SE	p	Naturali zed x Post Beta	SE	p	N
Medi-Cal	-0.000	0.006	0.98	-0.013	0.006	0.030	220,925
ED visit	+0.055	0.010	<0.001	+0.050	0.008	<0.001	220,925
Usual source	+0.054	0.012	<0.001	+0.007	0.008	0.37	220,925
Forgone care	-0.004	0.007	0.62	-0.012	0.005	0.027	220,925

Notes: Reference group: US-born citizens. All specifications include year FE and individual controls. Bold indicates $p < 0.05$.

Table 8. CHIS Adults Documentation demands

Coefficient	Estimate	SE	p-value	95% CI
Noncitizen	+0.204	0.007	<0.001	[0.191, 0.218]
Naturalized citizen	+0.122	0.004	<0.001	[0.113, 0.130]
Age	-0.000	0.000	<0.001	[-0.001, -0.000]
Female	+0.004	0.002	0.088	[-0.001, 0.009]
Below 200% FPL	+0.025	0.003	<0.001	[0.019, 0.031]

Notes: Outcome is an indicator for being asked for SSN or citizenship documentation when seeking health coverage. Reference group: US-born citizens. N = 220,925. Additional race/ethnicity controls included but not shown.

Appendix

Appendix Table A: COVID-19 Sensitivity - CHIS (Full Results)

Outcome	Specification	Beta (DiD)	SE	p-value	N
Medi-Cal	Drop COVID (20-21)	-0.003	0.006	0.61	146,774
ED visit	Drop COVID (20-21)	+0.064	0.011	<0.001	146,774
Usual source	Drop COVID (20-21)	+0.061	0.013	<0.001	146,774
Forgone care	Drop COVID (20-21)	+0.003	0.008	0.73	146,774
Medi-Cal	COVID interaction	-0.003	0.006	0.58	185,269
ED visit	COVID interaction	+0.064	0.011	<0.001	185,269
Usual source	COVID interaction	+0.061	0.013	<0.001	185,269
Forgone care	COVID interaction	+0.002	0.008	0.81	185,269

Appendix Table B: BRR Variance Estimation - CHIS (Full Results)

Outcome	Beta (DiD)	BRR SE	Robust SE	BRR t	BRR p	N replicates
Medi-Cal	-0.001	0.001	0.006	-1.63	0.10	80
ED visit	+0.055	0.001	0.010	38.42	<0.001	80
Usual source	+0.055	0.002	0.012	33.77	<0.001	80
Forgone care	-0.003	0.001	0.007	-2.66	0.008	80

Notes: BRR SE computed from 80 CHIS replicate weights (RAKEDW0-RAKEDW80). BRR standard errors are uniformly smaller than heteroskedasticity-robust standard errors.

Figures

Figure 1. ACS event study, Medicaid enrollment (full sample).

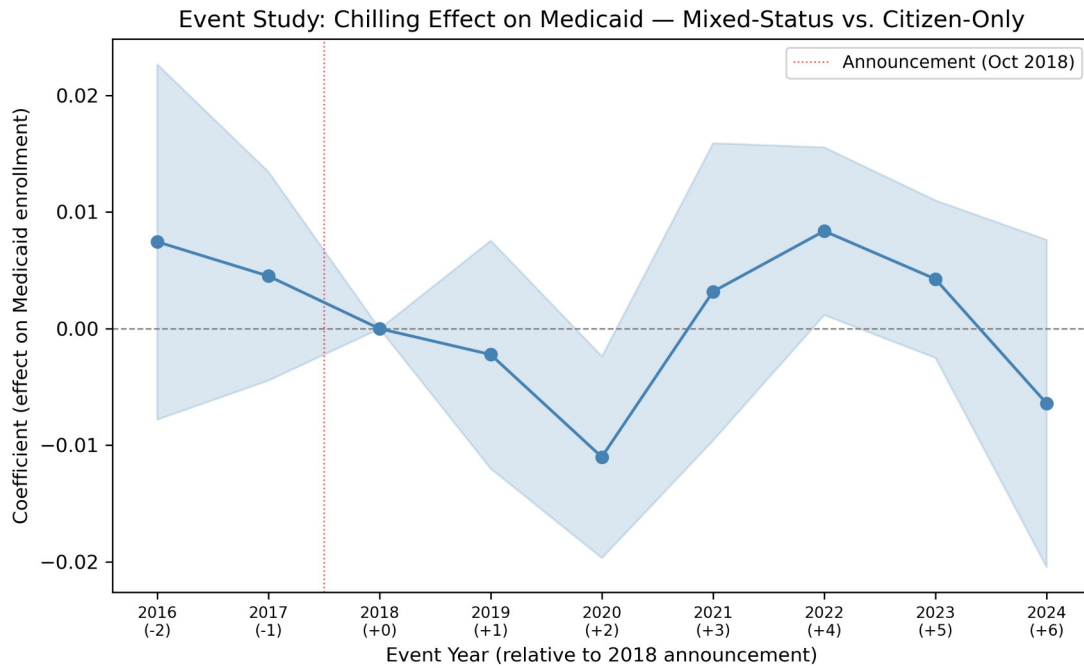


Figure 1

Event-study coefficients for the mixed-status x year interaction, 2018 reference year, with 95% confidence intervals based on state-clustered standard errors.

Figure 2. CHIS event studies, four outcomes.

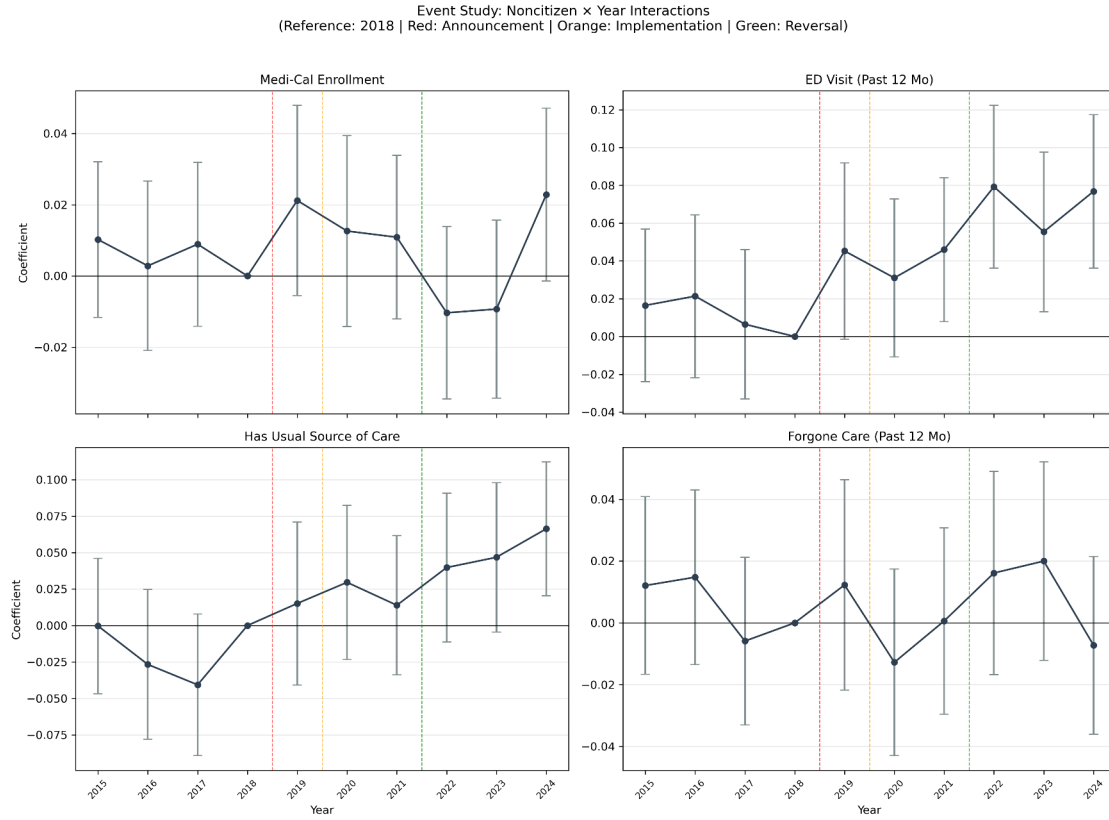


Figure 2

Event-study coefficients for the noncitizen x year interaction for Medi-Cal enrollment, ED visits, usual source of care, and forgone care, CHIS 2015-2024, 2018 reference year.

Figure 3. ACS event study, Medicaid enrollment, below 200% FPL.

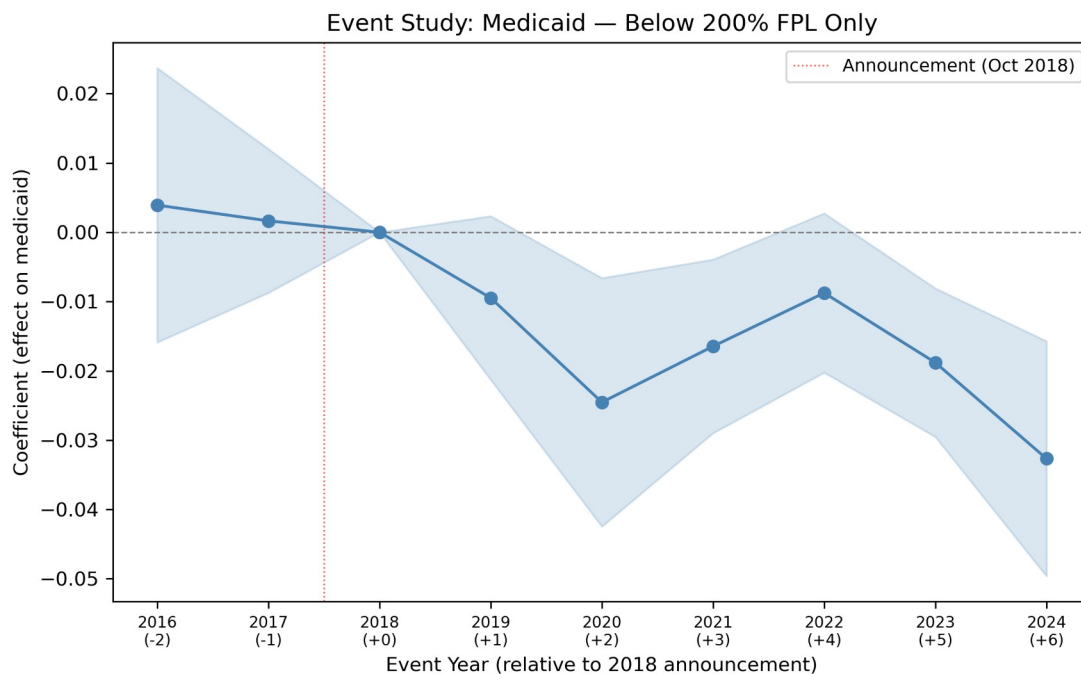


Figure 3

Event-study coefficients for the below-200% FPL sample, 2018 reference year.

Figure 4. Placebo event study, citizen-only families.

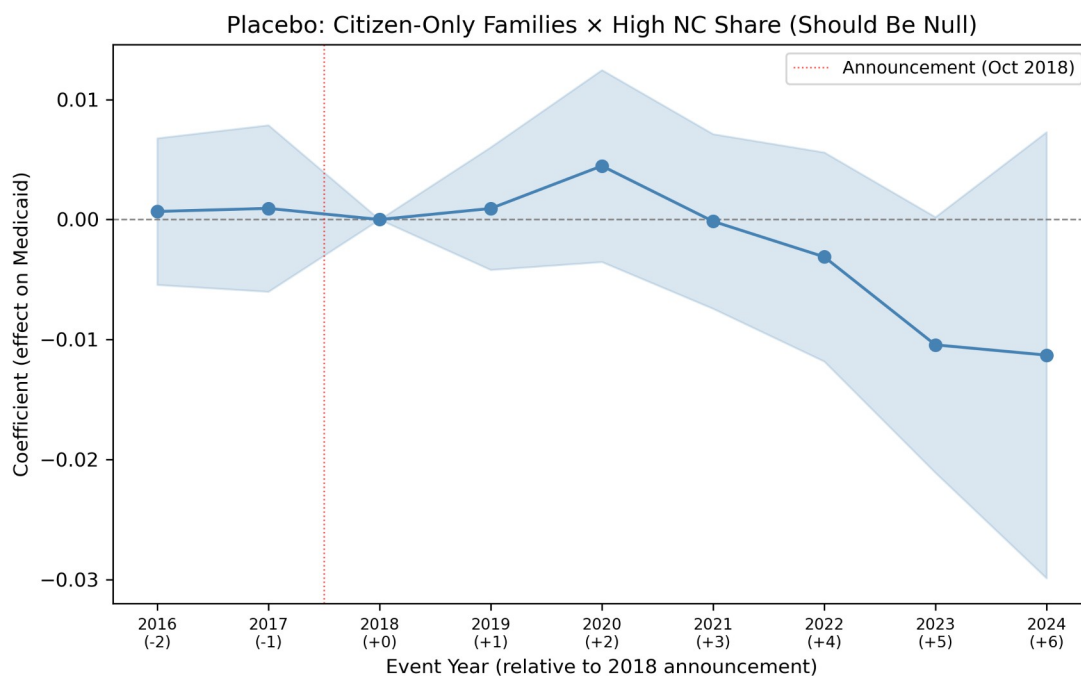


Figure 4

Placebo event-study coefficients for citizen-only families in high- versus low-noncitizen-share states; all coefficients are insignificant.

Figure 5. Pre-period covariate balance, ACS sample.

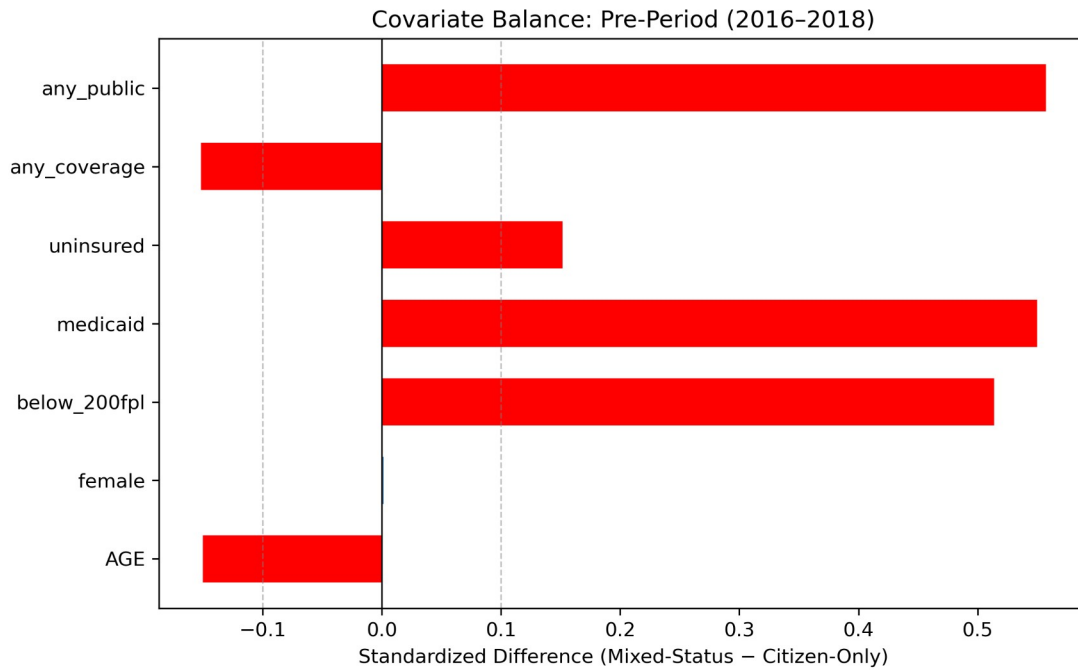


Figure 5

Pre-period differences between mixed-status and citizen-only children; level differences are absorbed by the family-type fixed effect.